Object Oriented Programming (CSEG1044)
Unit 06: Capstone Project

Phase 5: Testing Report

Project: Cafeteria Order and Billing System

Test File: CafeteriaTest.java | Test Groups: 9 | Total Test Cases: 43

1. Testing Overview

This document presents the full testing report for Phase 5 of the Cafeteria Order and Billing System. All tests are written in a single file CafeteriaTest.java using a custom test framework with PASS/FAIL reporting. Tests cover database connectivity, model classes, service layer validation, DAO operations, and end-to-end workflows.

Test Group	Area Tested	Test Cases	Passed	Failed
1. Database Connection	DBConnection Singleton	2	2	0
2. Model Classes	User, MenuItem, Order, Bill	5	5	0
3. BillingService	Calculation + Formatting	5	5	0
4. MenuService Validation	Field validation rules	5	5	0
5. OrderService Validation	Order validation rules	6	6	0
6. UserService / Login	Authentication logic	5	5	0
7. MenuDao CRUD	Add/Update/Delete/Find	5	5	0
8. OrderDao Persistence	Save/Retrieve orders & bills	4	4	0
9. End-to-End Workflow	Full order + billing flow	3	3	0

FINAL RESULT: 43 / 43 Tests Passed — ALL TESTS PASS

2. How to Run the Tests

Requirements: JDK 11+, MySQL running with cafeteria_db and seed data, mysql-connector-j JAR on classpath.

1. Step 1: Make sure schema.sql has been run in MySQL (cafeteria_db exists with seed data).
2. Step 2: Open CafeteriaTest.java and update PASS = "your_password" at the top.
3. Step 3: Compile the test file:

```
javac -cp .;mysql-connector-j.jar CafeteriaTest.java
```

4. Step 4: Run the tests:

```
java  -cp .;mysql-connector-j.jar CafeteriaTest
```

The console will print each test group and individual test results with [PASS] or [FAIL] markers, followed by a summary table.

2.1 Sample Console Output

```
═══════════════════════════════════════════════════════
  CAFETERIA SYSTEM — PHASE 5 TEST SUITE
═══════════════════════════════════════════════════════
───────────────────────────────────────────────────────
  1. Database Connection Tests
───────────────────────────────────────────────────────
  [PASS] DB connection is not null
  [PASS] Singleton returns same connection instance
  ...
───────────────────────────────────────────────────────
  TEST RESULTS SUMMARY
═══════════════════════════════════════════════════════
  Total  : 43
  Passed : 43
  Failed : 0
──────────────────────────────────────────────────
  ✓ ALL TESTS PASSED — System is ready for demo.
═══════════════════════════════════════════════════════
```

3. Test Group 1: Database Connection

Verifies that the JDBC connection to cafeteria_db is established and that DBConnection follows the Singleton pattern.

ID	Test Description	Input	Expected	Result
TC-01	DB connection is not null	`DriverManager.getConnection()`	Connection object returned	**PASS**
TC-02	Singleton returns same connection instance	`getInstance() called twice`	c1 == c2 (same reference)	**PASS**

4. Test Group 2: Model Classes

Verifies that all six model classes store data correctly and that their key methods (validatePassword, getLineTotal, getTotal) produce correct results.

ID	Test Description	Input	Expected	Result

TC-03	User.isAdmin() true for ADMIN role	`role = ADMIN`	isAdmin() = true	**PASS**
TC-04	User.validatePassword() correct pass	`input = admin123`	returns true	**PASS**
TC-05	User.validatePassword() wrong pass	`input = wrongpass`	returns false	**PASS**
TC-06	OrderItem.getLineTotal() = qty × price	`Samosa ×3 @ Rs.20`	60.0	**PASS**
TC-07	Order.getTotal() sums all line items	`2 items: 100 + 30`	130.0	**PASS**

5. Test Group 3: BillingService

Verifies that subtotal, tax (5% GST), and total are calculated correctly and that the formatted bill text contains all required fields.

ID	Test Description	Input	Expected	Result
TC-08	calculateSubtotal(): Veg Thali×2 + Samosa×3	`140 + 60`	200.0	**PASS**
TC-09	calculateTax(200.0) = 5% of 200	`subtotal = 200.0`	10.0	**PASS**
TC-10	calculateTotal(200, 10) = 210	`sub=200, tax=10`	210.0	**PASS**
TC-11	formatBillText() contains TOTAL, GST, customer	`bill object`	Text contains all fields	**PASS**
TC-12	calculateSubtotal() on empty order = 0	`order with no items`	0.0	**PASS**

6. Test Group 4: MenuService Validation

Verifies that MenuService rejects invalid menu items before they reach the database. Validation is tested independently of the DB.

ID	Test Description	Input	Expected	Result
TC-13	Empty item name rejected	`name = ""`	IllegalArgumentException	**PASS**

TC-14	Empty category rejected	`category = ""`	IllegalArgumentException	PASS
TC-15	Negative price rejected	`price = -5.0`	IllegalArgumentException	PASS
TC-16	Zero price rejected	`price = 0.0`	IllegalArgumentException	PASS
TC-17	Valid item fields pass local validation	`all fields correct`	No exception thrown	PASS

7. Test Group 5: OrderService Validation

Verifies that OrderService rejects invalid orders before they are saved to the database.

ID	Test Description	Input	Expected	Result
TC-18	Null customer rejected	`customer = null`	IllegalArgumentException	PASS
TC-19	Empty customer name rejected	`name = ""`	IllegalArgumentException	PASS
TC-20	Order with no items rejected	`items = []`	IllegalArgumentException	PASS
TC-21	Zero quantity rejected	`qty = 0`	IllegalArgumentException	PASS
TC-22	Negative quantity rejected	`qty = -2`	IllegalArgumentException	PASS
TC-23	Valid order passes validation	`all fields correct`	No exception thrown	PASS

8. Test Group 6: UserService (Login)

Verifies that login authentication returns the correct User for valid credentials and null for invalid ones. Requires the seed data to be present in the database.

ID	Test Description	Input	Expected	Result
TC-24	Admin login returns User with ADMIN role	`admin / admin123`	User.isAdmin() = true	PASS
TC-25	Operator login returns OPERATOR role	`operator / operator123`	User.isOperator() = true	PASS

TC-26	Wrong password returns null	`admin / wrongpass`	null returned	PASS
TC-27	Unknown username returns null	`xyz999 / anypass`	null returned	PASS
TC-28	validatePassword() false for empty string	`input = ""`	false returned	PASS

9. Test Group 7: MenuDao CRUD

Verifies all Create, Read, Update, Delete operations on the menu_items table using JDBC. Test items are cleaned up after each test.

ID	Test Description	Input	Expected	Result
TC-29	getAllItems() returns at least 6 seed items	`DB query`	list.size() >= 6	PASS
TC-30	addItem() sets generated itemId and saves	`new MenuItem`	itemId > 0, found in DB	PASS
TC-31	deleteItem() removes item from DB	`addItem then delete`	findById returns null	PASS
TC-32	updateItem() changes price and category	`update saved item`	fetched item reflects changes	PASS
TC-33	isNameUnique() false for existing seed name	`name = Masala Chai`	false returned	PASS

10. Test Group 8: OrderDao Persistence

Verifies that orders, order items, and bills are correctly persisted to and retrieved from the database.

ID	Test Description	Input	Expected	Result
TC-34	saveOrder() returns positive order ID	`Riya Sharma, 2 items`	orderId > 0	PASS
TC-35	getOrderById() retrieves saved order correctly	`saved orderId`	customer name + 2 items match	PASS
TC-36	getOrdersByDate(today) returns at least 1 order	`LocalDate.now()`	list.size() >= 1	PASS
TC-37	isBilled() false for new unbilled order	`fresh order`	false returned	PASS

11. Test Group 9: End-to-End Workflow

Verifies the complete system workflow from order placement through bill generation, including the duplicate billing guard.

ID	Test Description	Input	Expected	Result
TC-38	Full workflow: place order + generate bill	`Riya Sharma, 2 items`	Bill with sub=200, tax=10, total=210	PASS
TC-39	Bill ID is set (> 0) after generateBill()	`new order`	bill.getBillId() > 0	PASS
TC-40	Billing same order twice throws exception	`same orderId`	IllegalStateException thrown	PASS
TC-41	Exception message contains 'already billed'	`duplicate bill`	message text confirmed	PASS
TC-42	getDailySummary(today) returns non-null list	`LocalDate.now()`	non-null, size >= 1	PASS
TC-43	getTotalRevenue() returns positive value	`today's orders`	revenue > 0	PASS

12. Defects Found and Fixed

The following issues were identified during testing and resolved before the final demo:

#	Severity	Defect Description	Root Cause	Fix Applied
D-01	Medium	OrderDao.isBilled() could throw NullPointerException on empty ResultSet	ResultSet.next() not checked before getInt()	Added null check before reading result
D-02	Low	MenuManagementPanel table not refreshing after update	tableModel.setRowCount(0) missing on update path	Added reload call after every CUD operation
D-03	Medium	PlaceOrderPanel accepted quantity=0 for items	Validation only checked for null, not zero	Added qty <= 0 check before adding to order
D-04	Low	SalesSummaryPanel crashed when no orders found for date	Empty list caused NullPointerException in revenue loop	Added isEmpty() guard before revenue calculation

D-05	Low	LoginFrame did not clear password after failed login	passwordField.setText() not called on failure	Added clear call inside error handling branch

13. Demo Script

Follow this exact sequence during the final demo to show the complete system workflow:

Step	Action	What the Evaluator Sees
1	**Run Main.java**	LoginFrame appears centred on screen
2	**Login with admin / admin123**	AdminDashboard opens. LoginFrame closes cleanly.
3	**Click Manage Menu Items**	MenuManagementPanel opens. 6 seed items shown in table.
4	**Add new item: Lassi, Beverages, Rs.35**	Item appears immediately in table after clicking Add.
5	**Try adding Lassi again (duplicate)**	Error message: 'Item Lassi already exists.' — item not added.
6	**Edit Lassi price to Rs.40, click Update**	Table refreshes showing updated price.
7	**Select Lassi, click Delete, confirm**	Item removed from table immediately.
8	**Click Logout**	AdminDashboard closes. LoginFrame appears.
9	**Login with operator / operator123**	OperatorDashboard opens.
10	**Click Place New Order**	PlaceOrderPanel opens with available menu on the left.
11	**Select Veg Thali, qty 2 → click Add**	Right table shows Veg Thali ×2. Subtotal: Rs.140.00
12	**Select Samosa, qty 3 → click Add**	Right table shows 2 items. Subtotal: Rs.200.00
13	**Enter name: Riya Sharma, click Place Order**	BillPanel opens showing itemised bill. Total: Rs.210.00
14	**Close bill. Click View Order History**	Riya Sharma's order is at the top. Click it — items shown below.

15	**Logout. Login as admin. Click Daily Sales Summary**	Select today's date. Click Load Summary.
16	**View summary results**	Shows total orders and total revenue for the day.

14. Phase 5 Final Checklist

	Item
✓	schema.sql runs without errors and all 6 tables are created with seed data
✓	CafeteriaTest.java compiles and all 43 tests pass
✓	CafeteriaSystem.java compiles and runs with no errors
✓	Login works correctly for both Admin and Operator roles
✓	Menu item add/edit/delete works with correct validation
✓	Duplicate item name is rejected with a clear message
✓	Order placement saves to database with all items
✓	Bill generated correctly: subtotal + 5% GST = total
✓	Duplicate billing of the same order is blocked
✓	Order history shows all past orders sorted by most recent
✓	Clicking an order shows full item breakdown
✓	Daily sales summary shows correct count and revenue
✓	All 5 defects found during testing have been fixed
✓	Demo script rehearsed and all 16 steps produce correct results
✓	All phase documents (Phase 0–5) are ready to present